Assignment8 – Learning

Given: June 10   Due: June 15

Problem 8.1 (Neural Networks in Python)

Implement neural networks in Python by completing the implementation of `network.py` at `https://kwarc.info/teaching/AI/resources/AI2/network/`.

Hint: You can test your implementation with `test.py`. Note that `test_train_xor_gate` may occasionally fail for a correct solution because it is randomized.

Solution: See `https://kwarc.info/teaching/AI/resources/AI2/network/`.

Problem 8.2 (Statistical Learning)

We use two observations to determine if it has rained on our property: whether the ground is wet, and whether a bucket we left outside is full.

1. Model this situation as a naive Bayesian network with a boolean class and two boolean attributes.

   Solution: We use three boolean variables R (rain), G (ground wet), and B (bucket full) with edges $R \to G$ and $R \to B$.

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2. Explain why this network requires 5 parameters ($2n + 1$ where $n = 2$ is the number of attributes). Choose 5 names for the parameters and use them to give the conditional probability table of the network.

   Solution: We need

   - one parameter θ for the probability of the class variable R: $P(R = true) = \theta$ and $P(R = false) = 1 - \theta$
   - for each attribute, one parameter for each possible value of the class variable, i.e., 2 if the class variable is boolean:
     - θ_{G1} and θ_{G2} for attribute G: $P(G = true \mid R = true) = \theta_{G1}$ and $P(G = false \mid R = true) = 1 - \theta_{G1}$ as well as $P(G = true \mid R = false) = \theta_{G2}$ and $P(G = false \mid R = false) = 1 - \theta_{G2}$
     - θ_{B1} and θ_{B2} for attribute B: $P(B = true \mid R = true) = \theta_{B1}$ and $P(B = false \mid R = true) = 1 - \theta_{B1}$ as well as $P(B = true \mid B = false) = \theta_{B2}$ and $P(B = false \mid R = false) = 1 - \theta_{B2}$

3. Now assume we have observed for 50 days with the following results:

rain	ground wet	bucket full	number of days
yes	yes	yes	10
yes	yes	no	5
yes	no	yes	6
yes	no	no	4
no	yes	yes	2
no	yes	no	9
no	no	yes	3
no	no	no	11

State the formula for the likelihood of this list of 50 observations in terms of the 5 parameters.

Solution: Let **d** be the list of observations. The likelihood is

$$P(\mathbf{d} \mid h_{\theta,\theta_{G1},\theta_{G2},\theta_{B1},\theta_{B2}}) =$$

$$\theta^{10+5+6+4}(1-\theta)^{2+9+3+11}\theta_{G1}^{10+5}(1-\theta_{G1})^{6+4}\theta_{G2}^{2+9}(1-\theta_{G2})^{3+11}\theta_{B1}^{10+6}(1-\theta_{B1})^{5+4}\theta_{B2}^{2+3}(1-\theta_{B2})^{9+11}$$

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4. Give the Maximum Likelihood approximations for the 5 parameters given these 50 observations. (You just need to compute them, not derive the formula for computing them.)

Solution: We can derive the formulas by taking logarithms and maximizing. The resulting formulas are the fraction of positive over total examples. So we get

$$\theta = (10+5+6+4)/(10+5+6+4+2+9+3+11)$$

$$\theta_{G1} = (10+5)/(10+5+6+4) \qquad \theta_{G2} = (2+9)/(2+9+3+11)$$

$$\theta_{B1} = (10+6)/(10+5+6+4) \qquad \theta_{B2} = (2+3)/(2+9+3+11)$$

Problem 8.3 (Statistical Learning)

You observe the values below for 20 games of a sports team. You want to predict the result based on weather and opponent.

Weather	Opponent	Number of wins	losses
Rainy	Weak	3	1
Cloudy	Weak	0	1
Sunny	Weak	4	2
Rainy	Strong	0	2
Cloudy	Strong	2	3
Sunny	Strong	0	2

1. What is the hypothesis space for this situation, seen as an inductive learning problem?

   Solution: The set of functions $\{Rainy, Cloudy, Sunny\} \times \{Weak, Strong\} \to \{Win, Loss\}$.

2. Explain whether we can learn the function by building a decision tree.

   Solution: We cannot. Even all attributes together, i.e., $Weather$ and $Opponent$, do not determine the result. So no decision tree exists.

3. To apply Bayesian learning, we model this situation as a Bayesian network $W \to R \leftarrow O$ using random variables W (weather), O (opponent), and R (game result). What are the resulting entries of the conditional probability table for the cases
   1. $P(W = rainy) =$ $\boxed{3/10}$
   2. $P(R = win \mid O = weak) =$ $\boxed{7/11}$

   Solution: $P(W = rainy) = 3/10$ and $P(R = win \mid O = weak) = 7/11$

Problem 8.4 (Passive Reinforcement Learning)

Consider the example on Passive Learning in 4×3 world from the slides.

1. Give the transition model to the extent that it can be learned from these trials.

   Solution: To obtain $P(s' \mid s, a)$, we count how often s' followed s after executing a. Concretely, we obtain
   - $P((1,2) \mid (1,1), Up) = 2/3$ and $P((2,1) \mid (1,1), Up) = 1/3$
   - $P((1,3) \mid (1,2), Up) = 3/3$

- $P((2,3) \mid (1,3), Right) = 2/3$ and $P((1,2) \mid (1,3), Right) = 1/3$
- $P((3,1) \mid (2,1), Left) = 1/1$
- $P((3,3) \mid (2,3), Right) = 2/2$
- $P((3,2) \mid (3,1), Left) = 1/1$
- $P((3,3) \mid (3,2), Up) = 1/2$ and $P((4,2) \mid (3,3), Up) = 1/2$
- $P((3,2) \mid (3,3), Right) = 1/3$ and $P((4,3) \mid (3,3), Right) = 2/3$

The transition probabilities for any action in state (4,1) and for other action in the other states are not learned.

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2. How could we learn the entire model?

Solution: Because we use a fixed policy, we always apply the same a whenever reaching s. So only some parts of the transition model can be learned from these trials. To learn the entire model, we have to try all actions in each state.

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3. How would we proceed to learn the utilities of the states?

Solution: Once we have learned the transition model, we can apply the Bellman equation to obtain a system of linear equations in the utilities of each state and solve that. (If we are only interested in the utilities of states relative to a fixed policy, we only need the partial transition model learned from trials with that policy.)

Problem 8.5 (Active Reinforcement Learning)

Consider reinforcement learning in an unknown non-deterministic environment.

1. Explain the difference between a passive and an active agent.
2. What is the critical trade-off in designing an actively learning agent?

Solution:

1. A passive agent has a fixed policy and is only trying to learn its utility. An active agent is additionally trying to find the best policy.
2. Because the optimal policy depends on the transition model, the agent must first learn the transition model and then find the optimal policy. But it is difficult to decide when to switch from the former to the latter. If it switches too early, the transition model has not been learned well yet and (garbage in, garbage out) the computed policy is bad. If it switches too late, it wastes time and resources learning the transition model in areas that are not visited by the optimal policy anyway.